



Program	BACHELOR OF TECHNOLOGY (B.Tech)	Semester - 5
Type of Course	Professional Elective Course	
Prerequisite	Basic knowledge of Python programming, data structures, algorithms, and elementary statistics.	
Rationale	Data analytics helps organizations extract meaningful insights from data for effective decision-making and business growth. This course focuses on industry-oriented analytics workflow including data acquisition, cleaning, transformation, exploratory analysis, visualization, dashboard development, and AI-assisted analytics. It enables students to analyse real-world datasets, communicate insights through interactive visualisations and dashboards, and apply data-driven techniques using modern analytics and business intelligence tools.	
Effective From A.Y.	2026-27	

Teaching Scheme (Contact Hours)				Examination Scheme				
Lecture	Tutorial	Lab	Credit	Theory Marks		Practical Marks		Total Marks
				SEE TH	IAT	SEE P	CCE	
3	0	2	4	70	30	50	50	200

SEE - Semester End Examination, IAT - Internal Assessment Test, CCE - Continues & Comprehensive Evaluation

Course Content		T - Teaching Hours W - Weightage	
Sr.	Topics	T	W
1	Introduction to Data Analytics - Data - Type of Data - Sources of Data - Data Quality and Data-Driven Decision Making - Introduction to Data Analytics - Importance of Data Analytics - Data Analytics vs Business Intelligence vs Data Science - Data Analytics Lifecycle - Types of Analytics: Descriptive Analytics, Diagnostic Analytics, Predictive Analytics, Prescriptive Analytics - Methods of Data Analytics	7	10
2	Data Acquisition and Preprocessing - Introduction to NumPy Arrays - Introduction to Pandas DataFrames - Data Import from CSV, Excel and JSON - Data Inspection and Exploration - Data Cleaning Techniques - Handling Missing Values - Duplicate Data Handling - Data Transformation - Data Aggregation - Filtering and Sorting - GroupBy Operations - Data Export	8	20
3	Exploratory Data Analysis and Statistical Techniques	8	20



Course Content		T - Teaching Hours W - Weightage	
Sr.	Topics	T	W
	• Exploratory Data Analysis (EDA) • Descriptive Statistics • Measures of Central Tendency • Measures of Dispersion • Frequency Distribution • Quartiles and Percentiles • Correlation Analysis • Covariance • Outlier Detection • Distribution Analysis • Identifying Trends and Patterns • Insight Generation from Data		
4	Data Visualization • Principles of Effective Data Visualization • Data Storytelling • Visualization Best Practices • Introduction to Matplotlib • Introduction to Seaborn • Line Charts • Bar Charts • Histograms • Scatter Plots • Box Plots • Pie Charts • Heatmaps • Pair Plots • Selecting Appropriate Visualization Techniques	8	20
5	Interactive Visualization and Dashboard Development • Introduction to Interactive Data Visualization • Plotly Express • Plotly Graph Objects • Dashboard Design Principles • Key Performance Indicators (KPIs) • Interactive Charts • Introduction to Dash Framework • Developing Interactive Dashboards • Business Reporting and Visualization	8	20
6	Business Analytics Applications, Ethics and Case Studies • Analytics in Business Decision Making • Applications of Data Analytics in Healthcare, Banking and Finance, Retail and E-Commerce, Manufacturing, Education • Social Media Analytics • Data Privacy and Ethics • Bias in Data Analytics • Case Studies using Open Datasets • Interpretation and Communication of Analytical Results	6	10
Total		45	100

**Suggested Distribution Of Theory Marks Using Bloom's Taxonomy**

Level	Remembrance	Understanding	Application	Analyze	Evaluate	Create
Weightage	15	25	25	15	10	10

NOTE : This specification table shall be treated as a general guideline for the students and the teachers. The actual distribution of marks in the question paper may vary slightly from above table.

Course Outcomes

At the end of this course, students will be able to:

CO1	Explain the concepts of data analytics, analytics lifecycle, types of analytics, and the role of data-driven decision making in solving engineering and business problems.
CO2	Acquire, clean, transform, and preprocess datasets using Python libraries to prepare data for analytical tasks.
CO3	Perform exploratory data analysis using descriptive statistical techniques to identify trends, relationships, and patterns in datasets.
CO4	Develop effective data visualizations using Python libraries to represent and communicate analytical insights.
CO5	Design interactive dashboards and business reports using modern visualization tools for effective decision support.
CO6	Evaluate data analytics solutions by interpreting results, considering ethical issues, and applying analytics in real-world domains.

CO PO Mapping

CO	CO - 1	CO - 2	CO - 3	CO - 4	CO - 5	CO - 6
PO - 1						
PO - 2						
PO - 3						
PO - 4						
PO - 5						
PO - 6						
PO - 7						
PO - 8						
PO - 9						
PO - 10						
PO - 11						

**Reference Books**

1.	Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani, An Introduction to Statistical Learning with Applications in Python, Springer, 2023
2.	Cole Nussbaumer Knaflic, Storytelling with Data: Let's Practice!, Wiley, 2021.
3.	Alberto Cairo, The Truthful Art: Data, Charts, and Maps for Communication, New Riders, 2016.
4.	Joel Grus, Data Science from Scratch: First Principles with Python, 2nd Edition, O'Reilly Media, 2019
5.	Allen B. Downey, Think Stats: Exploratory Data Analysis in Python, 2nd Edition, O'Reilly Media, 2014
6.	Wes McKinney, Python for Data Analysis: Data Wrangling with Pandas, NumPy, and Jupyter, 3rd Edition, O'Reilly Media, 2022. ISBN: 978-1098104030 (TextBook)
7.	Jake VanderPlas, Python Data Science Handbook: Essential Tools for Working with Data, 2nd Edition, O'Reilly Media, 2023. ISBN: 978-1098121228 (TextBook)
8.	Claus O. Wilke, Fundamentals of Data Visualization, O'Reilly Media, 2020. ISBN: 978-1492031086 (TextBook)

List of Practical

1.	Study the Data Analytics lifecycle and perform data acquisition from CSV, Excel, and JSON files using Pandas.
2.	Perform data inspection, cleaning, handling missing values, duplicate removal, and data transformation using Pandas.
3.	Apply filtering, sorting, aggregation, and GroupBy operations on real-world datasets.
4.	Perform Exploratory Data Analysis (EDA) using descriptive statistics and identify trends in datasets.
5.	Analyze datasets using correlation, covariance, outlier detection, and distribution analysis.
6.	Create basic data visualizations (Line, Bar, Histogram, Scatter, Pie).
7.	Develop advanced visualizations (Box Plot, Heatmap, Pair Plot, Violin Plot).
8.	Create interactive visualizations using Plotly Express and Plotly Graph Objects.
9.	Design and develop a simple interactive dashboard using Dash/Plotly for business reporting.
10.	Mini Project: Analyze a real-world dataset, build an interactive dashboard, and present actionable insights with ethical considerations.